INVERSE LAPLACE TRANSFORM: TWO-STAGE RC NETWORK

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Derive the transfer function of a loaded two-stage RC network and find its poles.	☐
2	Use partial fractions and the inverse Laplace transform to predict the step response, and relate each pole to a part of the response.	☐
3	Measure the step response with the Arduino Uno and compare it with the prediction, including the effect of loading between the stages.	☐

2.0 Hardware Set-Up

This practical exercise builds on the RC circuit used in the practical exercise, RC Circuit Charging: ODE and Time Solution, and uses the Arduino Uno and breadboard from the ACE-Lab Base + Sense kit with two resistors and two capacitors. Values of $R_1 = R_2 = 100$ kΩ and $C_1 = C_2 = 10$ µF are used in the example.

Connect a digital output pin (for example D7) through R_1 to node 1, with C_1 from node 1 to GND. Connect node 1 through R_2 to node 2, with C_2 from node 2 to GND. Measure the voltage at node 2 on an analogue input (for example A0). For the optional unloaded test, an op-amp voltage follower is placed between node 1 and R_2.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The transfer function of the network is derived, its step response is found with the inverse Laplace transform, and the prediction is compared with measurements.

The exercise supports the ACE-CORE learning guide, Inverse Laplace Transform.

Derive the Transfer Function

Writing Kirchhoff's current law at nodes 1 and 2 in the s-domain, and eliminating the voltage at node 1, gives the transfer function from the input voltage to the voltage at node 2:

$$\frac{V_2(s)}{V_{in}(s)} = \frac{1}{R_1C_1R_2C_2\,s^2 + (R_1C_1 + R_2C_2 + R_1C_2)\,s + 1}$$

The term R_1C_2 appears because the second stage draws current from the first, known as loading. Without it, the network would simply be two first-order stages in series. With the example values, all three time products are 1 s, so:

$$\frac{V_2(s)}{V_{in}(s)} = \frac{1}{s^2 + 3s + 1}, \qquad p_{1,\,2} = \frac{-3 \pm \sqrt{5}}{2} = -0.382, \quad -2.618$$

Find the Step Response

For a 5 V step, $V_{in}(s) = 5/s$. Expanding in partial fractions gives one term for the step and one for each pole:

$$V_2(s) = \frac{5}{s\,(s - p_1)(s - p_2)} = \frac{A_0}{s} + \frac{A_1}{s - p_1} + \frac{A_2}{s - p_2}$$

$$v_2(t) = 5\,(1 - 1.171\,e^{\,-0.382t} + 0.171\,e^{\,-2.618t})\ \mathrm{V}$$

Each pole contributes an exponential term. The slow pole at −0.382 s^{-1} has a large residue and dominates the response. The fast pole at −2.618 s^{-1} has a small residue and only shapes the first second, giving the S-shaped start.

1. Derive the transfer function by hand from Kirchhoff's current law at the two nodes.
2. Calculate the residues A_0, A_1 and A_2 by hand, and check them with the MATLAB function `residue`.

Measure the Step Response

1. Use the model from the practical exercise, RC Circuit Charging: ODE and Time Solution, with a Pulse Generator period of 30 s so that the output is high for 15 s.
2. Discharge both capacitors, then run the model and log the voltage at node 2 and the time.
3. Run the script below to compare the measurement with the prediction.
4. Optionally, add the op-amp voltage follower between node 1 and R_2, repeat the test and compare it with the unloaded prediction $1/(s + 1)^2$.

MATLAB Script: predicting the step response with residue and comparing it with the measurement

```
v_k = v.signals.values(:);  t_k = t.signals.values(:);
R1 = 100e3; R2 = 100e3; C1 = 10e-6; C2 = 10e-6;
den = [R1*C1*R2*C2, R1*C1 + R2*C2 + R1*C2, 1];
[res, p] = residue(5, conv(den, [1 0]))      % residues and poles of V2(s)

tt = t_k(t_k <= 15);
v_pred = real(res.' * exp(p*tt.'));          % sum of residue*exp(pole*t)
v_unl  = 5*step(tf(1, conv([R1*C1 1], [R2*C2 1])), tt);  % no loading

plot(tt, v_k(t_k <= 15), 'k.', tt, v_pred, 'b', tt, v_unl, 'r--'); grid on;
xlabel('Time [s]'); ylabel('Voltage at node 2 [V]');
legend('Measured', 'Loaded prediction', 'Unloaded (two separate stages)', ...
       'Location', 'southeast');
```

The predicted responses for the example components are shown in Figure 1. The loaded network is noticeably slower than two separate stages would be, and after about 2 s the response is almost exactly the slow pole term alone.

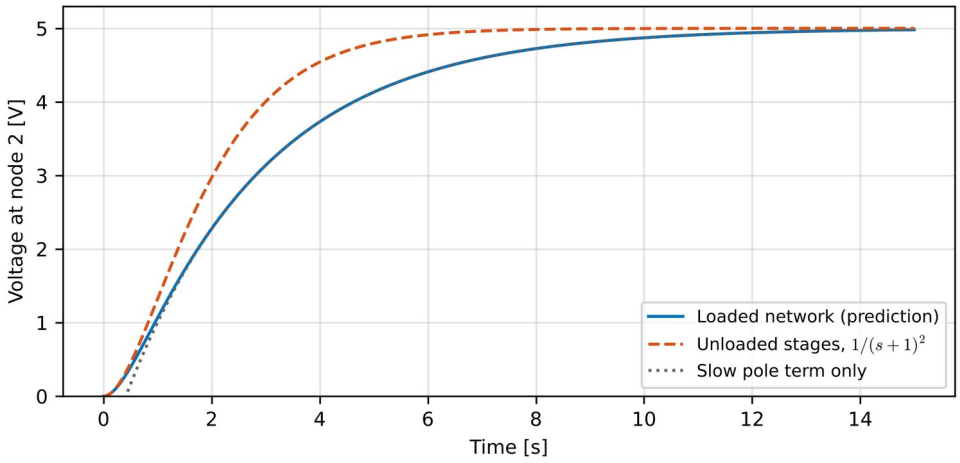


Figure 1: *Theoretical Step Response of the Two-Stage RC Network for R = 100 kΩ and C = 10 µF*

Table 2: *Results for Your Two-Stage Network*

Quantity	Predicted	Measured
Voltage at *t* = 1 s (V)	1.07	
Voltage at *t* = 3 s (V)	3.14	
Time to reach 2.5 V (s)	2.23	
Overshoot	none	

1. Explain why a passive RC network like this one can never overshoot.
2. Which pole would you keep in a simplified first-order model of the network, and why? Write that model.

3. How do the sizes of the residues show how much each pole contributes to the response?

4.0 Concluding Remarks

This practical exercise predicted the step response of a two-stage RC network with the inverse Laplace transform and compared it with measurements. Using MATLAB, Simulink and ACE-Lab hardware, the transfer function was derived, its poles were found, and partial fractions turned the s-domain response into a sum of exponentials.

Each pole contributed one exponential term, and the slow pole with the large residue dominated the response. The loading between the stages slowed the network compared with two independent stages. Reading the behaviour of a system from its poles and residues, and simplifying it by keeping only the dominant pole, are used throughout control design.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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